

Keshav Dahiya

Curriculum Vitae

Department of Mathematics,
Indiana University - Indianapolis,
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Research Interests

Quantum Algebras, Representation Theory, and Integrable Systems

Education

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| 2021 - 2025 | Ph.D. in Mathematics
Purdue School of Science, Indianapolis
Advisor: Evgeny Mukhin
Thesis: <i>Intertwiners of representations of quantum affine algebras and superalgebras</i> |
| 2018 - 2020 | M.S. in Mathematics
Indian Institute of Technology, Delhi
Advisor: Biplab Basak
Thesis: <i>Triangulation of manifolds</i> |
| 2013 - 2015 | M.Tech. in Information and Communication Technology
Indian Institute of Technology, Delhi
Advisor: Brejesh Lall
Thesis: <i>Signal reconstruction from its noisy sub-band components</i> (Link) |
| 2010 - 2015 | B.Tech. in Electrical Engineering
Indian Institute of Technology, Delhi |

Preprints and Publications

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| 2025 | K. Dahiya and E. Mukhin, <i>Intertwiners of representations of quantum affine superalgebras</i> , in preparation |
| 2025 | K. Dahiya and E. Mukhin, <i>Intertwiners of representations of twisted quantum affine algebras</i> , J. Math. Phys. 66 (5) , 2025 (Editor's Pick)
DOI arXiv |
| 2025 | K. Dahiya and E. Mukhin, <i>Intertwiners of representations of untwisted quantum affine algebras and Yangians revisited</i> , J. Math. Phys. 66 (5) , 2025 (Editor's Pick)
DOI arXiv |

Teaching Experience

Instructor of Record

Indiana University, Indianapolis, IN

Fall 2025 MATH-I-110: Fundamentals of Algebra
Summer 2025 MATH-I-165: Analytic Geometry and Calculus II
Spring 2025 MATH-I-165: Analytic Geometry and Calculus I

Purdue University, Indianapolis, IN

Fall 2024 MA 16010: Applied Calculus I

Indiana University - Purdue University (IUPUI), Indianapolis, IN

Spring 2024 MATH 166-00: Analytic Geometry and Calculus II
Fall 2023 MATH 166-00: Analytic Geometry and Calculus II
Summer 2023 MATH 110-00: Fundamentals of Algebra
Spring 2023 MATH 171-00: Multidimensional Mathematics
Fall 2022 MATH 165-00: Analytic Geometry and Calculus I

Recitation Leader

Indiana University - Purdue University (IUPUI), Indianapolis, IN

Spring 2022 MATH 166-00: Analytic Geometry and Calculus II

Teaching Assistant

Indian Institute of Technology, Delhi

Fall 2014 EEL 205: Signals and Systems

Invited Talks

Oct 2025 **Mathematical Physics Seminar**
Purdue University, West Lafayette, IN

Oct 2024 **Quantum Groups and Representation Theory Conference**
North Carolina State University, Raleigh, NC

Mar 2024 **14th Southeastern Lie Theory Conference**
University of Virginia, Charlottesville, VA

Seminar Presentations

Representation Theory Seminar

Indiana University, Indianapolis, IN

Mar 2025 *The quantum determinant for Yangian of $\mathfrak{gl}_n$*
Mar 2025 *Transfer matrices for Yangian of $\mathfrak{gl}_n$*
Feb 2025 *Fusion of R matrices*

Oct/Nov 2024	<i>MacDonald polynomials</i>
Sep 2024	<i>Howe duality and quantum Weyl group</i>
Dec 2023	<i>Crystal bases for $U_q\mathfrak{sl}_n$ and Young tableaux</i>
Jan/Feb 2023	<i>The $\mathfrak{gl}_m \times \mathfrak{gl}_n$ duality</i>
Sep 2022	<i>The Pfaffian using Grassmann variables</i>
Mar/Apr 2022	<i>The Gelfand-Tsetlin algebras and representations of symmetric groups</i>
Dec 2021	<i>Characters of representations of finite groups</i>

K-Theory Seminar

Indiana University, Indianapolis, IN

May/June 2024	<i>Generalized cohomology of complex vector bundles</i>
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Graduate Student Seminar

Indiana University, Indianapolis, IN

Apr 2023	<i>Quantum computing: Shor's algorithm</i>
Oct 2022	<i>Tilings by regular vertices</i>
Mar 2022	<i>Geometric group theory: Quasi-isometry, the Svarc-Milnor lemma</i>
Dec 2021	<i>Spectral bounds in graph theory</i>
Mar 2021	<i>Discrete Fourier transform, the convolution theorem and FFT algorithm</i>

Professional Service

2024	Journal Reviewer , <i>Transformation Groups</i> , Volume 29
2023	Journal Reviewer , <i>Journal of Mathematical Physics</i> , Volume 64

Projects

Fall 2025	<i>Image Caption Generator</i> (GitHub) The Erdős Institute, Deep Learning Bootcamp
Summer 2025	<i>Quantum Hamming code</i> (GitHub) The Erdős Institute, Quantum Computing Bootcamp
Summer 2025	<i>Quantum state preparation</i> (GitHub) The Erdős Institute, Quantum Computing Bootcamp
Summer 2025	<i>Quantum circuit for constrained optimization</i> (GitHub) The Erdős Institute, Quantum Computing Bootcamp
Spring 2022	<i>q-characters of quantum affine algebras</i> (GitHub) Advisor: Evgeny Mukhin, Purdue School of Science, Indianapolis
Fall 2012	<i>Captcha recognition</i> Advisor: Sumeet Agarwal, Indian Institute of Technology, Delhi

Fall 2012	<i>3-D object rendering using ray tracing</i> Advisor: Subodh Kumar, Indian Institute of Technology, Delhi
Spring 2012	<i>Guidance system for visually impaired</i> Advisor: Ranjan Bose, Indian Institute of Technology, Delhi

Professional Experience

2015 - 2018	Ed-Tech Business Founder Delhi, India • Built a web application using PHP and JavaScript on a private domain to deliver virtual lessons to IIT aspirants. • Designed and taught interactive online Mathematics and Physics lessons integrating problem-solving exercises.
Summer 2013	Computer Vision Intern Qualcomm, Hyderabad, India • Developed Android applications for real-time edge detection, corner detection, face detection and contour generation in the mobile camera video using FastCV SDK. • Designed user interfaces in Java and implemented native Android functionality in C++ on top of FastCV SDK.

Awards and Achievements

2024	Outstanding Advanced Mathematics Graduate Student Award Indiana University, Indianapolis
2021	Outstanding Beginning Mathematics Graduate Student Award Purdue School of Science, Indianapolis
2021 - 2022	Graduate Student Scholarship Purdue School of Science, Indianapolis
2010	National Rank - 332 among 0.5 million participants (99.93 percentile) Joint Entrance Examination for Engineering Indian Institutes of Technology
2010	National Rank - 296 among 1 million participants (99.97 percentile) All India Engineering Entrance Examination Central Board of Secondary Education, India
2009	National Olympiad Rank - 72 among 50k participants (99.86 percentile) National Talent Search Examination National Council of Educational Research and Training, India

- 2006 **State Rank - 1**
 Maths Mind Competition
 Mount Abu Public School, Delhi
- 2006 **State Rank - 5**
 Ramanujan Mathematics Competition
 New State Academy, Delhi

Referees

Evgeny Mukhin

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Department of Mathematical Sciences,
Indiana University, Indianapolis
Email: emukhin@iu.edu
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Daniel Ramras

Associate Professor of Mathematics
Department of Mathematical Sciences,
Indiana University, Indianapolis
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Vitaly Tarasov

Professor of Mathematics
Department of Mathematical Sciences,
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Phone: (317) 274-8144

Roland Roeder

Professor (Director, Graduate Programs)
Department of Mathematical Sciences
Indiana University, Indianapolis
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